

QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int num1 = sc.nextInt();
6         int num2 = sc.nextInt();
7         int sum = num1 + num2;
8         int difference = num1 - num2;
9         int product = num1 * num2;
10        System.out.println("Sum: " + sum);
11        System.out.println("Difference: " + difference);
12        System.out.println("Product: " + product);
13        if (num2 != 0) {
14            double quotient = (double) num1 / num2;
15            System.out.println("Quotient: " + quotient);
16        } else {
17            System.out.println("Quotient: Undefined (division by zero)");
18        }
19    }
20 }
```

OUTPUT

```
Sum: 13
Difference: 7
Product: 30
Quotient: 3.3333333333333335
```

INPUT

```
10
3
```

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int a = sc.nextInt();
6         int b = sc.nextInt();
7         int c = sc.nextInt();
8         int max;
9         if (a >= b && a >= c) {
10             max = a;
11         } else if (b >= a && b >= c) {
12             max = b;
13         } else {
14             max = c;
15         }
16         System.out.println("Maximum: " + max);
17     }
18 }
```

OUTPUT

Maximum: 27

INPUT

17
13
27

QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class leap{
3     public static void main(String[] args){
4         int year;
5         Scanner sc = new Scanner(System.in);
6         year=sc.nextInt();
7         if((year%400==0)|| (year%4==0 && year%100!=0))
8             System.out.print(year+" is a Leap year");
9         else
10            System.out.print(year+" is not a Leap year");
11     }
12 }
```

INPUT

2024

OUTPUT

```
2024 is a Leap year
```


QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;  
2 class fib{  
3     public static void main(String[] args){  
4         Scanner sc=new Scanner(System.in);  
5         int n=sc.nextInt();  
6         int a=0,b=1,c;  
7         for(int i=1;i<=n;i++){  
8             System.out.print(a+" ");  
9             c=a+b;  
10            a=b;  
11            b=c;  
12        }  
13    }  
14 }
```

INPUT

9

OUTPUT

0 1 1 2 3 5 8

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class factorial{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int num = sc.nextInt();
6         int fact=1;
7         for(int i=1;i<=num;i++){
8             fact*=i;
9         }
10        System.out.print("Factorial of "+ num +" is "+ fact);
11    }
12 }
```

INPUT

3

OUTPUT

```
Factorial of 3 is 2
```

QUESTIONS 31 / 31 completed

RT-Pattern ▾

RT-PATTERN

Write a Java program to print the following pattern using loops:*

```
**  
***  
****  
*****
```

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;  
2 class rtpattern{  
3     public static void main(String[] args){  
4         Scanner sc = new Scanner(System.in);  
5         int rows=sc.nextInt();  
6         for(int i=1;i<=rows;i++){  
7             for(int j=1;j<=i;j++){  
8                 System.out.print("* ");  
9             }  
10            System.out.println();  
11        }  
12    }  
13 }
```

INPUT

5

OUTPUT

```
*  
* *  
* * *  
* * * *  
* * * * *
```

QUESTIONS 31 / 31 completed

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class BinarySearch {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         int n = sc.nextInt();
6         int[] arr = new int[n];
7         for (int i = 0; i < n; i++) {
8             arr[i] = sc.nextInt();
9         }
10        int target = sc.nextInt();
11        int result = binarySearch(arr, target);
12        if (result == -1) {
13            System.out.println("Element not found in the array.");
14        } else {
15            System.out.println("Element found at index: " + result);
16        }
17        sc.close();
18    }
19    public static int binarySearch(int[] arr, int target) {
20        int low = 0;
21        int high = arr.length - 1;
```

INPUT

```
5
4
3
2
1
6
6
```

OUTPUT

```
Element found at index: 4
```


QUESTIONS 31 / 31 completed

GCD ▾

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2 class GCD{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int a,b;
6         a=sc.nextInt();
7         b=sc.nextInt();
8         while(b!=0){
9             int temp = b;
10            b = a%b;
11            a = temp;
12        }
13        System.out.print("GCD = "+a);
14    }
15 }
```

INPUT

12
36

OUTPUT

GCD = 12

QUESTIONS 31 / 31 completed

Prime ▼

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java ▼

Run

Save

```
1 import java.util.Scanner;
2 class prime{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         int num = sc.nextInt();
6         boolean isPrime = true;
7         if(num>1){
8             for(int i=2;i<num;i++){
9                 if(num%i==0){
10                     isPrime = false;
11                     break;
12                 }
13             }
14         } else {
15             isPrime = false;
16         }
17         if(isPrime)
18             System.out.print("Prime");
19         else
20             System.out.print("Not Prime");
21     }
```

OUTPUT

Prime

INPUT

7

QUESTIONS 31 / 31 completed

Sort ▾

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java ▾

Run

Save

```
1 import java.util.Scanner;
2 public class SelectionSort {
3     public static void selectionSort(int[] arr) {
4         int n = arr.length;
5         for (int i = 0; i < n - 1; i++) {
6             int minIndex = i;
7             for (int j = i + 1; j < n; j++) {
8                 if (arr[j] < arr[minIndex]) {
9                     minIndex = j;
10                }
11            }
12            if (minIndex != i) {
13                int temp = arr[i];
14                arr[i] = arr[minIndex];
15                arr[minIndex] = temp;
16            }
17        }
18    }
19    public static void main(String[] args) {
20        Scanner sc = new Scanner(System.in);
21        int n = sc.nextInt();
```

INPUT

5
3
5
9
2
1

OUTPUT

Sorted array: 1 2 3 5 9